

EXPERIMENT 27: PROLOG PROGRAM FOR FRUIT AND ITS COLOR USING BACKTRACKING

Aim

To write and execute a Prolog program to represent fruits and their colors and retrieve the information using backtracking.

Objective

- To understand facts and predicates in Prolog.
- To represent relationships between fruits and colors.
- To understand the concept of backtracking.
- To retrieve multiple solutions for a query.

Algorithm

1. Define facts representing different fruits and their colors.
2. Create a predicate to represent the fruit-color relationship.
3. Enter a query for a particular fruit or color.
4. Prolog searches the knowledge base for a matching fact.
5. If multiple matches are available, Prolog uses backtracking to return the next solution.
6. Display all matching results.

Flowchart

START

|

v

Define Fruit-Color Facts

|

v

Enter Query

|

v

Search Knowledge Base

|
v
Mat
ch
Fou
nd?

/\
Yes No

| |
v v
Display False

Answer

|
v
More Solutions?

/\
Yes No

| |
v v

Backtrack STOP

Code

% Fruit and Color using Backtracking

```
fruit(apple, red). fruit(banana,  
yellow). fruit(orange,  
orange). fruit(grapes, green).  
fruit(mango, yellow).  
fruit(guava, green).
```

```
fruit(strawberry, red).
```

```
fruit(watermelon, green).
```

```
color_of_fruit(Fruit, Color) :-
```

```
fruit(Fruit, Color).
```

Query

```
?- fruit(Fruit, yellow).
```

Output

```
Fruit = banana ; Fruit
```

```
= mango ; false.
```

Result

The program successfully identifies fruits and their colors using Prolog facts and backtracking.

Conclusion

Thus, fruit-color relationships were successfully implemented using Prolog and the concept of backtracking.

```
% Fruit and Color using Backtracking
```

```
fruit(apple, red).  
fruit(banana, yellow).  
fruit(orange, orange).  
fruit(grapes, green).  
fruit(mango, yellow).  
fruit(guava, green).  
fruit(strawberry, red).  
fruit(watermelon, green).
```

```
color_of_fruit(Fruit, Color) :-  
    fruit(Fruit, Color).
```

```
3 ?- make.
```

```
% c:/users/varshini m/onedrive/desktop/her  
true.
```

```
4 ?- color_of_fruit(mango, Color).  
Color = yellow.
```

```
5 ?- █
```